Teacher's notes

OBJECTIVES

- Increase a quantity by a given percentage.
- Decrease a quantity by a given percentage.
- Solve problems relating to discounts.
- Solve problems relating to percentage composition.
- Calculate the percentage profit or loss when an item is sold.

PRIOR KNOWLEDGE

Students should be able to perform basic calculations with percentages. They should also be able to calculate a percentage of a quantity and express one quantity as a percentage of another.

BACKGROUND

Most of these activity sheets have been developed for students to complete without using a calculator. It may be appropriate for students to complete some activity sheets using pen and paper methods and then check their answers using a calculator. This will reinforce both pen and paper and calculator skills.

When calculators are being used, it is important that students write at least one line of working to show the calculation that they have performed.

Some students may need assistance with the subtleties of the English language when solving problems involving percentages. For example, *10% of $40* is **not** the same as *10% off $40*.

STARTER ACTIVITY

Hold a brief mental maths quiz, challenging students to give equivalent fractions for common percentage and vice versa; for example, 10%, 50%, 25%. Then lead a terminology discussion to help students think about whether an answer is going to be more or less than the given quantity. For example, '$100 is increased by 10%. Will the answer be more or less than $100?' Review terms such as *increase by, decrease by, discount, off, of, marked up by, marked down by*.

ACTIVITY SHEETS

2.1 Percentage increase

This activity sheet develops students' skills in increasing a quantity by a given percentage. Two methods are shown in the example. You should direct each student to **one** of these methods, depending on the student's level of competency.

Answers

1 a $72 b $180 c $276 d 900 mL e 96 kg f 5.76 m

2 a $55 b $224 c $54.60 d $210 e $793.50 f $31.35 g 112 km h 63 tonnes i 3.68 kg j $135

3 $149.50

4 $100.80

5 $762.20

2.2 Percentage decrease

This activity sheet develops students' skills in decreasing a quantity by a given percentage. Two methods are shown in the example. You should direct each student to **one** of these methods, depending on the student's level of competency.

Answers

1 a $72 b $117 c $58.50 d 180 km e 405 L f 8.82 kg

2 a $176 b $102 c $255 d $942 e $516 f $17.10 g 66 m h 128 kg i 59.5 L j 175 km

3 16 928

4 $1 295

5 $626.25

2.3 Discounts

In this activity sheet, students will apply their skills in decreasing by a percentage to problems involving discounts.

Answers

1 a $16 b $10 c $60 d $36 e $3 f $42

2 a $54, $486 b $23.50, $211.50 c $379, $1 516 d $64.50, $1 225.50 e $54.90, $494.10 f $129, $731

3 $36.50

4 a $56 b $224

5 $117.30

6 $1 775.55